

02.04.2026

$$\bar{X}_{CG} - \bar{X}_{AC} [\%] \rightarrow 5\% \dots 10\%$$

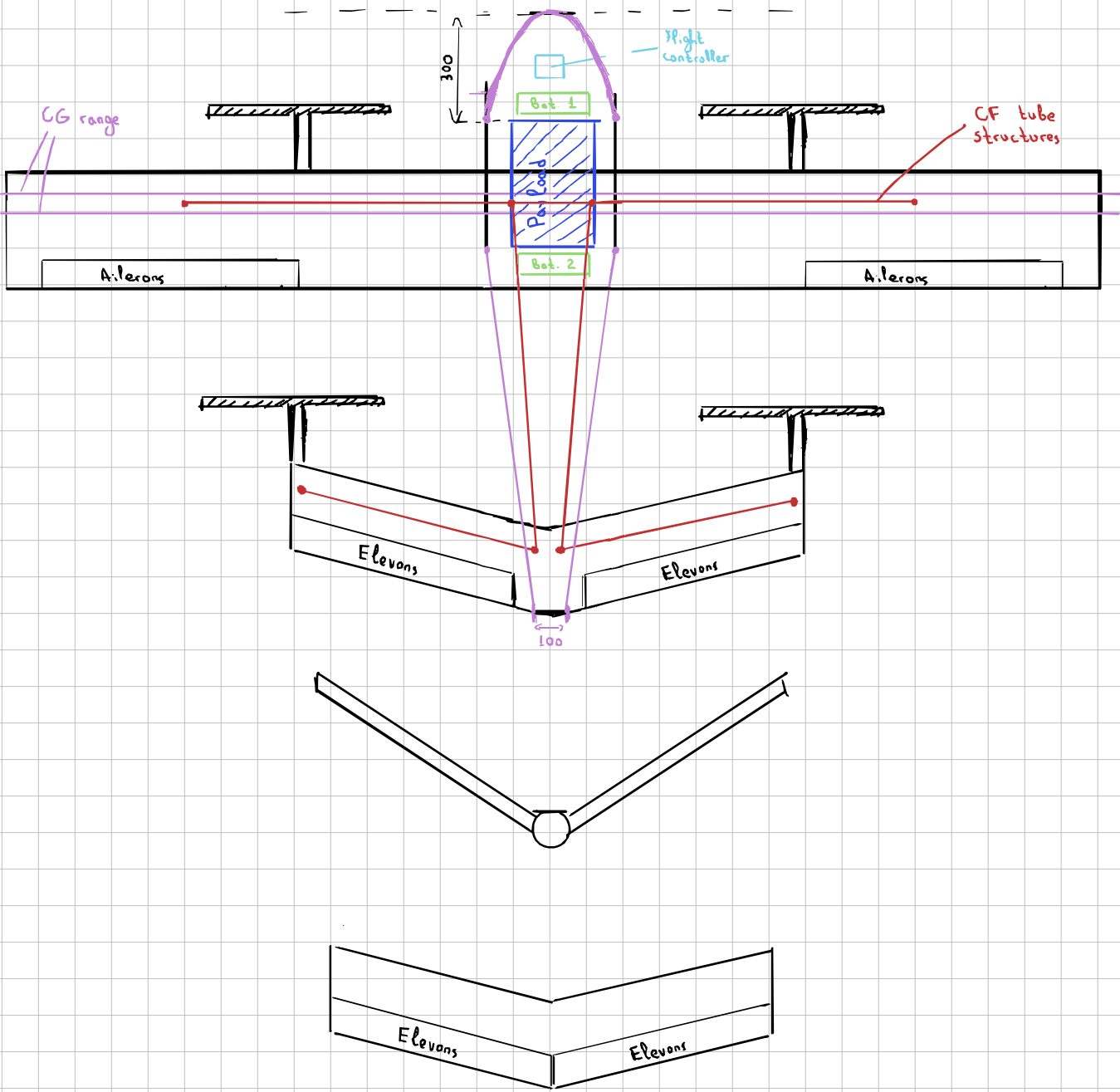
↳ For stable aircraft, the stability measure is negative

$$\bar{X}_{AC} = \frac{C_{Ma,w} - C_{Ma,fus} + C_{Ma,hr}}{C_{Ld,w} + C_{Ld,hr}}$$

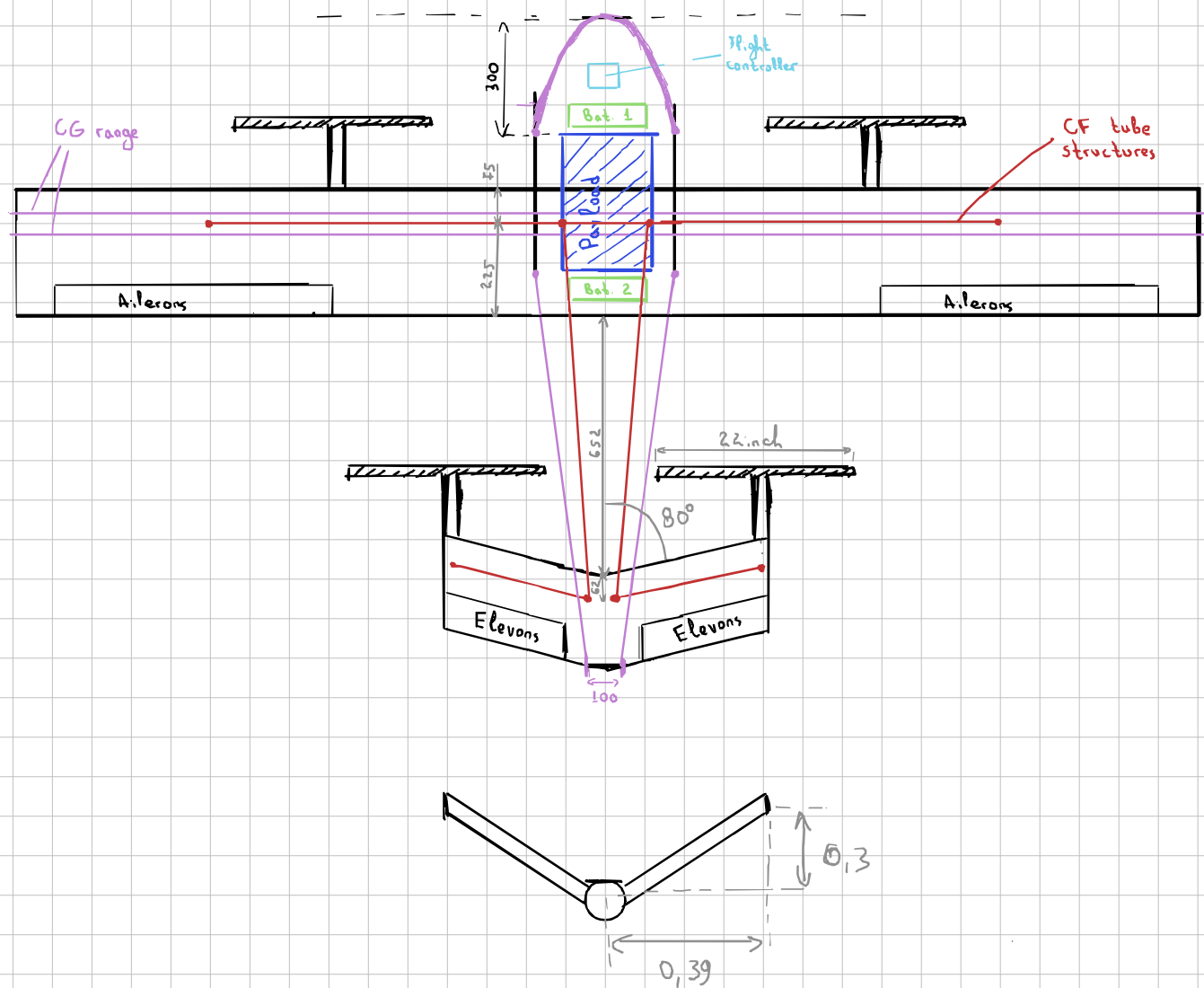
$$\bar{X}_{AC,w} = \frac{X_{0,w} + \overset{\%}{\left(n_{AC,w} \cdot C_{root,w} \right)}}{C_{MAC}}$$

$$C_{Ld,w} = \frac{2\pi \cdot AR}{2 + \sqrt{\frac{AR^2 \beta^2}{k^2} \cdot (1 + 0)} + 4}$$

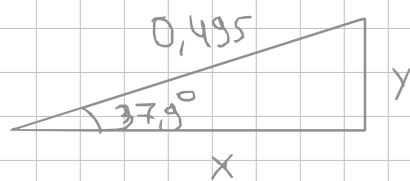
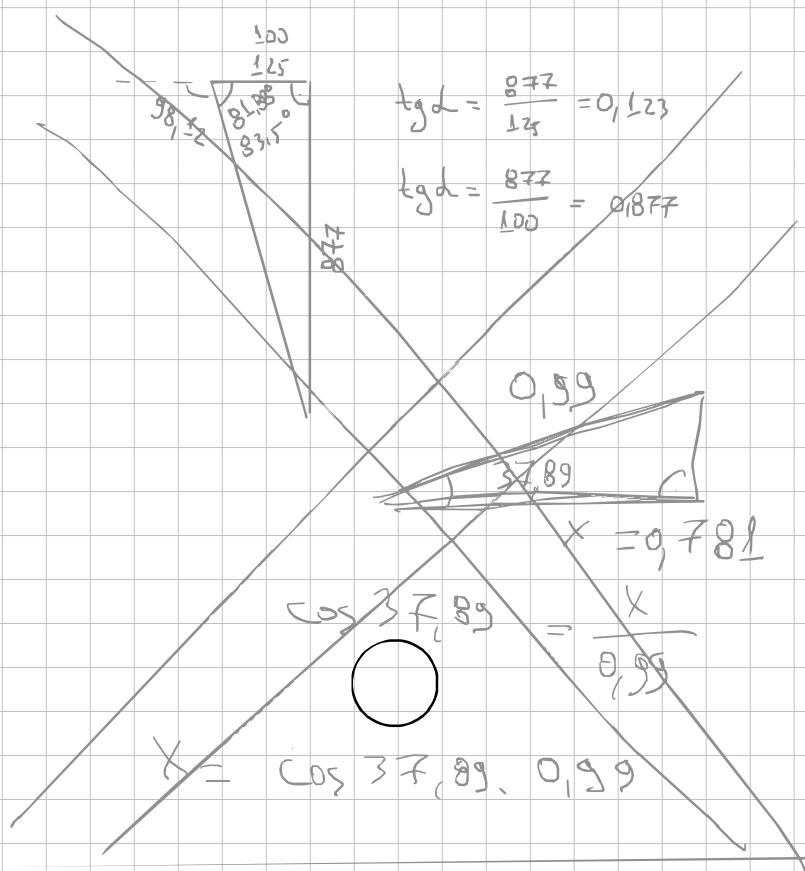
09.04.2026



Last Update : 01. May. 2026



Not correct!



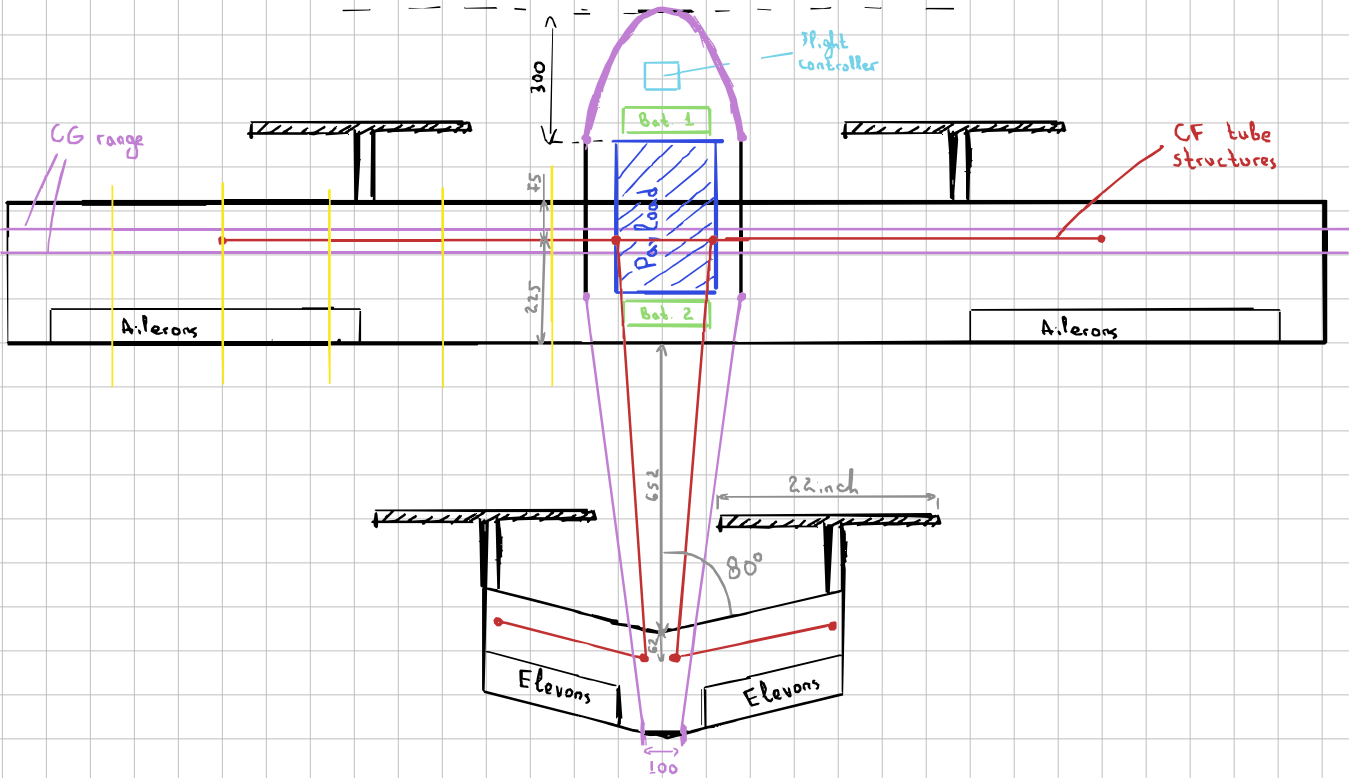
$$\cos 37.9 = \frac{x}{0.495}$$

$$x = 0.789 \cdot 0.495 = \underline{\underline{0.39}}$$

$$\sin 37.9 = \frac{y}{0.495}$$

$$y = \underline{\underline{0.304}}$$

Last Update: 01.05.2026

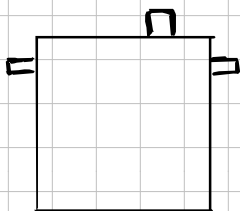


Not correct!

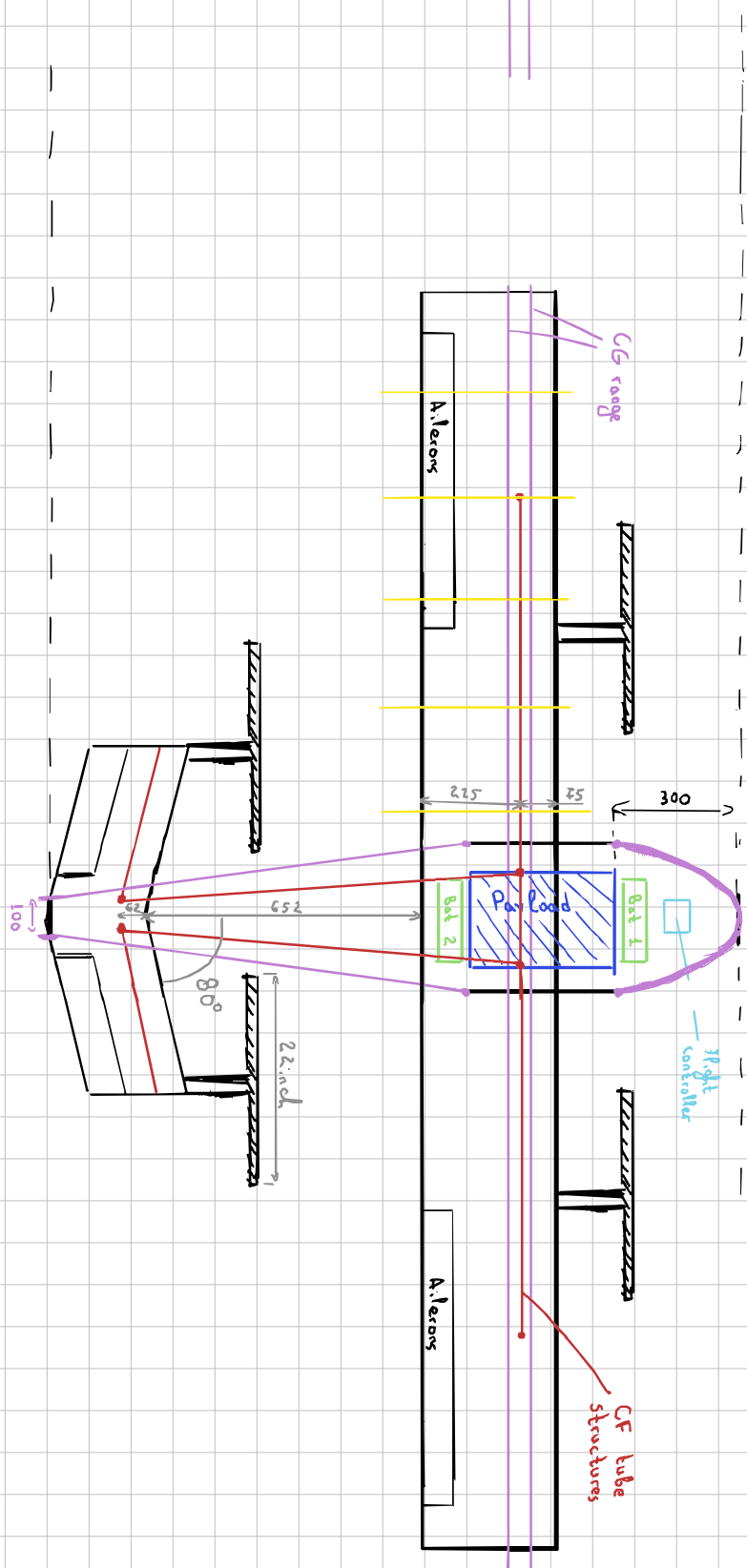
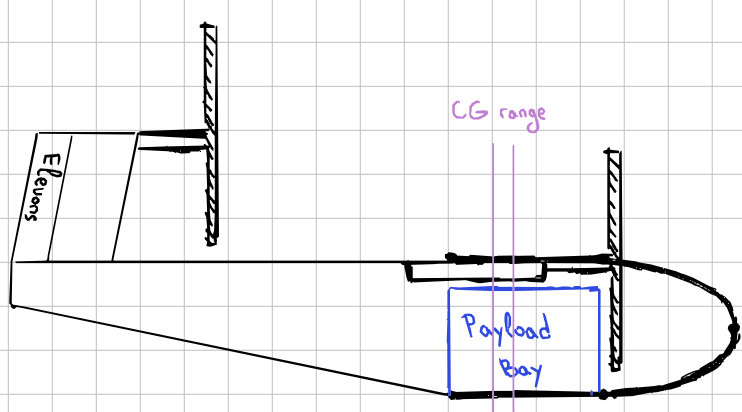
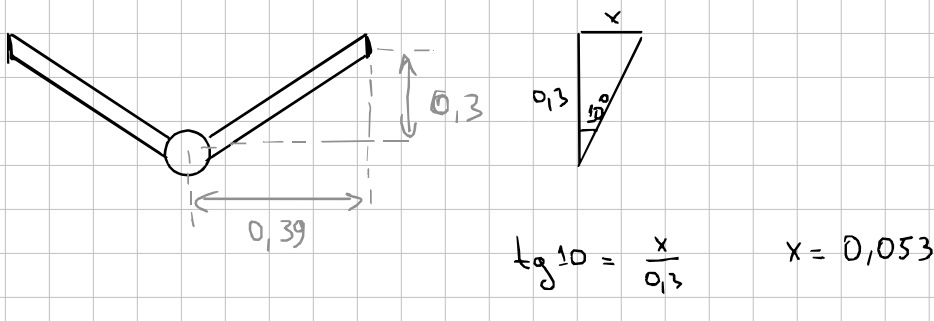
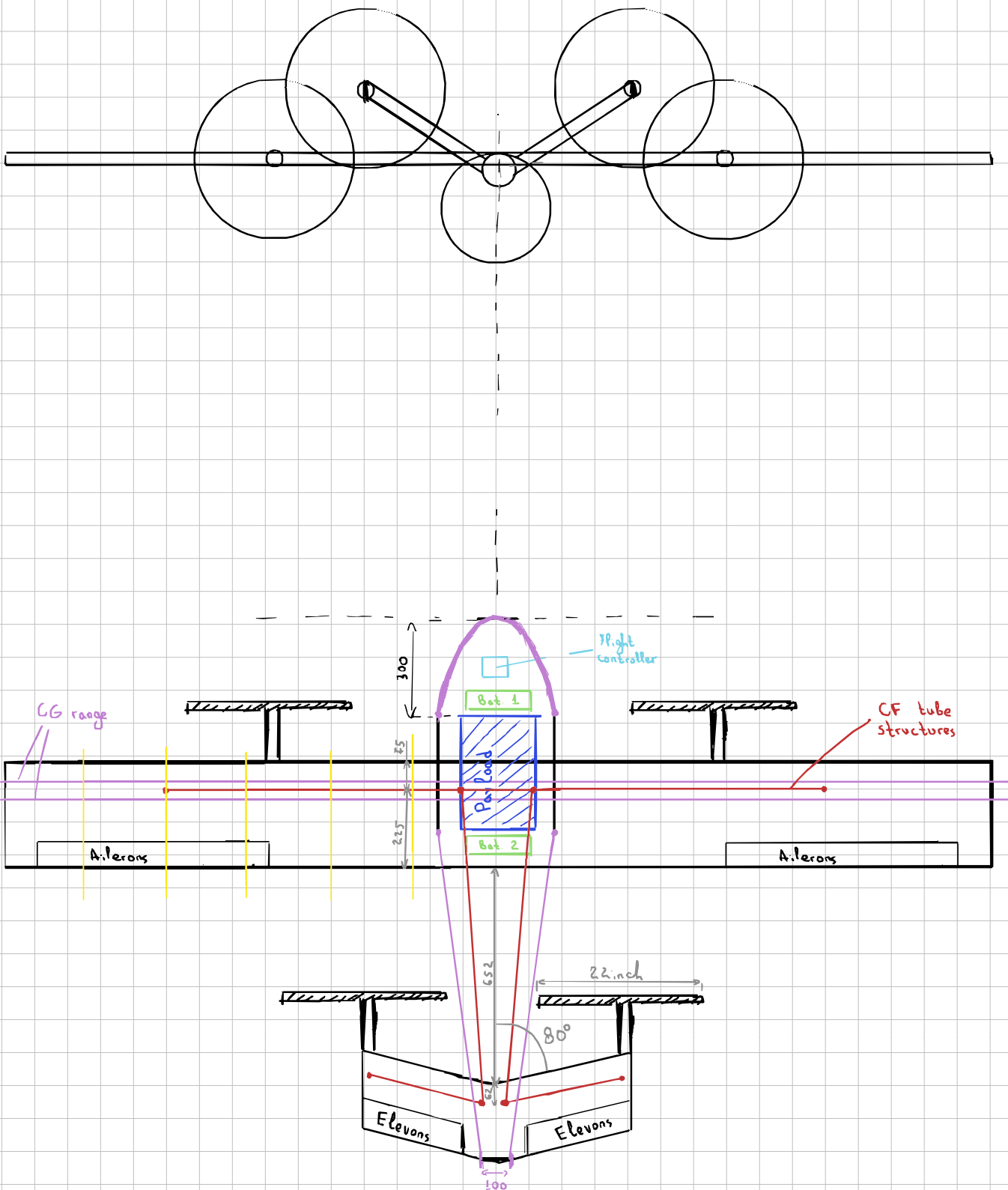
$\tan \alpha = \frac{877}{125} = 0,123$
 $\tan \alpha = \frac{877}{100} = 0,877$
 $\alpha = 37,89^\circ$
 $x = 0,781$
 $\cos 37,89 = \frac{x}{0,99}$
 $x = \cos 37,89 \cdot 0,99$

13.04.2026

Tilt Mechanism Concept:

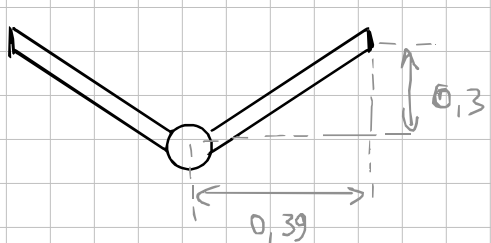
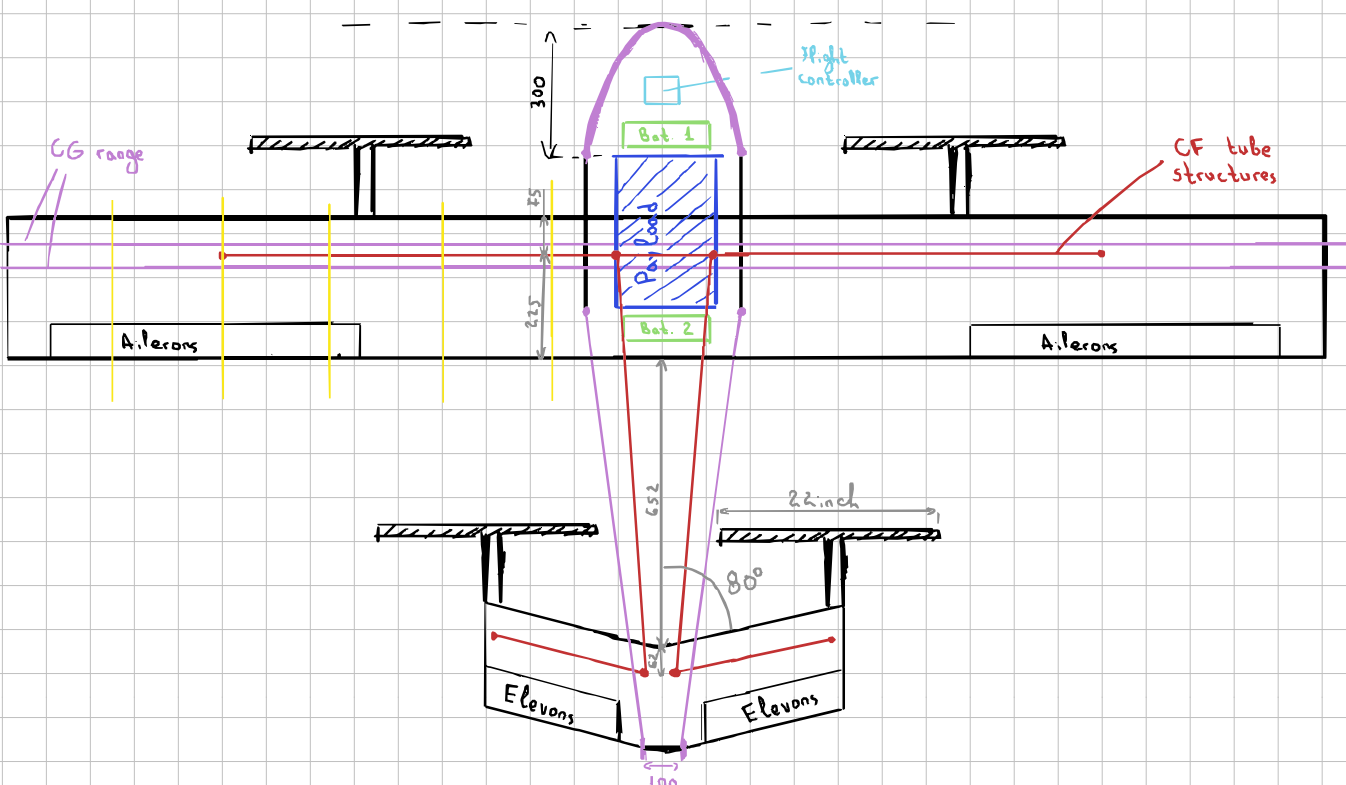
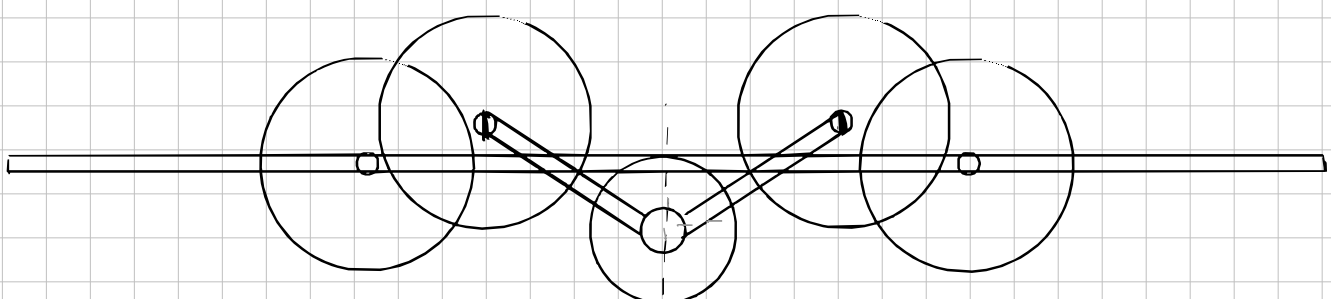


01.05.2026



02.05.2026

Updated low V-tail position w.r.t. the fuselage



$$\tan 10 = \frac{x}{0,3}$$

$$x = 0,053$$

